

LUIS CARLOS HERNANDEZ

Technical Engineering Leader | Principal Engineer | Engineering Manager
Enterprise Architecture | Java | Cloud Platforms | APIs | DevOps | AI-Assisted Engineering
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PROFESSIONAL SUMMARY

Technical engineering leader with 20+ years of experience designing, building, modernizing, and supporting enterprise software platforms across healthcare, financial services, aerospace, defense, transportation, retail, and eCommerce. Combines deep hands-on engineering background with proven leadership of distributed software teams, technical programs, production systems, and enterprise delivery initiatives.

Experienced in Java/J2EE, Spring, REST/SOAP services, cloud platforms, distributed systems, SQL databases, DevOps, CI/CD, production support, and engineering process improvement. Known for translating complex business and system requirements into scalable architectures, reliable delivery plans, and maintainable software solutions.

Currently focused on technical leadership, engineering intelligence, AI-assisted development practices, production support improvement, release governance, and operational visibility for enterprise software organizations.

TECHNICAL LEADERSHIP HIGHLIGHTS

- 20+ years delivering enterprise software across healthcare, financial services, aerospace, defense, transportation, retail, and eCommerce.
- Led technical design, architecture, development, testing, integration, and production support for mission-critical systems.
- Managed and mentored engineering teams across the United States, Mexico, and India while remaining engaged in architecture, design reviews, delivery planning, and technical decisions.
- Delivered Java, J2EE, Spring, REST/SOAP, Tomcat, Hibernate, SQL, cloud, and embedded software solutions across regulated and high-reliability environments.
- Supported software modernization, cloud adoption, API design, CI/CD, testing automation, production support, and engineering operations.
- Built engineering intelligence and dashboard-driven approaches to improve delivery quality, production support, and decision-making.

CORE TECHNICAL COMPETENCIES

Architecture & Design: Enterprise Architecture, Distributed Systems, API Design, SOA, Microservices, Data Modeling, Integration Architecture, Non-Functional Requirements, System Design, Technical Roadmaps

Engineering & Development: Java, Java/J2EE, Spring, REST APIs, SOAP Services, JSON, XML, JSP, Servlets, JavaBeans, Hibernate, C, C++, C#.NET, Python, Shell Scripting, JavaScript, PHP, SQL

Cloud & DevOps: AWS, EC2, ECS, Lambda, S3, DynamoDB, DocumentDB, Aurora, CloudWatch, Jenkins, Git, GitHub, GitLab, CI/CD, Docker, Tomcat, Linux, Unix

Leadership & Delivery: Technical Leadership, Engineering Management, Agile Delivery, SDLC, Production Support, Release Management, Cross-Functional Collaboration, Vendor Coordination, Mentoring, Requirements Analysis, QA/Test Planning

Tools & Methods: Jira, Confluence, MS Project, DOORS, ClearCase, Subversion, CVS, jMeter, Swagger, OAuth2, MISRA, DO-178B, DO-254

PROFESSIONAL EXPERIENCE

Academic HealthPlans / Brown & Brown - Manager, Applications Development | 2024 - Present | Healthcare Technology

- Provide technical and operational leadership for application development teams supporting healthcare platforms, production issues, release readiness, and cross-functional delivery.
- Drive production support transformation by improving intake, triage, prioritization, ownership, escalation paths, and visibility across engineering workstreams.
- Build engineering intelligence dashboards and operational reporting to improve leadership visibility into capacity, feature delivery, quality, production support, and engineering effort allocation.
- Support enterprise release governance, regression planning, testing coordination, vendor alignment, and stakeholder communication for high-risk releases.
- Contribute to AI-enabled engineering initiatives focused on production support triage, engineering analytics, executive dashboards, and reusable intelligence patterns.

Cognizant - Senior Engineering Manager | 2018 - 2024 | Enterprise Consulting - Apple, Mastercard, Canada Life

- Led engineering teams supporting multiple Apple business units, including EHS Incidents, Chemicals, Inspections, Industrial Hygiene, AppleCare, Retail, and IS&T.
- Coordinated development, testing, technical discussions, release planning, customer communication, and vendor alignment across onshore and offshore teams.
- Managed 40 engineers across Mexico and multiple Scrum teams delivering enterprise financial technology capabilities for the Mastercard Send Partner ecosystem.
- Supported architecture and implementation using Java 11, Oracle, Spring, REST/SOAP services, JSON, JUnit, Maven, Angular, Tomcat, Redis, and Swagger.
- Led discovery and planning for an Android/iOS insurance application, including stakeholder interviews, architecture discussions, DevOps considerations, testing strategy, API planning, and implementation roadmap.

Digital Room Inc. - Manager, Software Engineering | 2020 - 2021 | eCommerce / AWS / Printing Platform

- Led approximately 10 software engineers across backend, frontend, operations, database, and internal customer workstreams.
- Designed and delivered cloud-enabled systems using AWS Lambda, S3, DynamoDB, DocumentDB, Aurora, EC2/ECS, OAuth2, CloudWatch, and REST APIs.
- Designed a Watchdog health monitoring mechanism integrated with AWS CloudWatch and standardized service health visibility across applications.
- Supported modernization from legacy PHP architecture toward cloud-based services, improved monitoring, and scalable software delivery practices.

Leonardo DRS - Manager, Software Engineering | 2018 - 2020 | Defense / Embedded Systems

- Led software architecture, requirements, interface design, coding, testing, and documentation for embedded systems using C, C++, C#, firmware, and formal engineering practices.

- Developed schedules, metrics, software development plans, requirements specifications, software descriptions, and software test plans.
- Managed staffing forecasts, development execution, formal qualification testing, process compliance, technical reviews, and quality standards.

General Electric Transportation - Technical Lead Software Engineer | 2015 - 2018 | *Transportation / Locomotive Systems*

- Led technical operations for requirements capture, high-level design, implementation, testing, and integration of ECU application software using Java.
- Designed and implemented seven Java-based modules running in containerized environments, including CMS, Onboard Maintenance, Speed Module, GPS, CAB, EXC, and AUX modules.
- Supported migration of legacy software to a Predix platform using embedded Tomcat, Spring, hypervisor-based architecture, and Hibernate.
- Developed test harnesses using Java, Python, and shell scripting and created JSP, servlet, and JavaBean components for simulation and testing.

Eaton Aerospace - Lead Software Engineer | 2010 - 2015 | *Aerospace / Simulation / Test Systems*

- Led software architecture, requirements, interface design, development, testing, and formal qualification activities using C, C++, Java, Python, and Linux/Tomcat environments.
- Implemented aircraft system simulations using Java/J2EE, Apache Tomcat, C++, Java, and Python interfaces.
- Developed software requirements, design documents, test plans, and qualification support for aerospace validation systems.
- Supported Jenkins-based continuous integration for regression testing of simulation software.

EARLIER ENGINEERING EXPERIENCE

Crane Aerospace - Senior Software Engineer | 2007 - 2010: Developed Java/J2EE, Hibernate, JDBC, C++, Python, and Perl components for A400M military aircraft anti-lock braking simulation and spent six months on customer-site integration in Madrid, Spain.

Astronautics of America - Lead Software Engineer | 2006 - 2007: Led development of Java/Spring/Tomcat applications and test harness capabilities for Airbus A400M aircraft systems, including Jenkins release mechanisms and Jython hardware interfaces.

Rockwell Collins - Software Engineer | 2004 - 2006: Developed DO-178B Level C avionics software in C/C++ on LynxOS RTOS, including CMS maintenance systems, configuration algorithms, UDP communication, and pilot checklist functionality.

Lockheed Martin / Atmos Energy - Software Engineer | 2003 - 2004: Built Java, MySQL, Hibernate, Linux, and web-based internal applications for inventory and data acquisition systems.

United States Army / 82nd Airborne Division - Weapons Sergeant | 2001 - 2003: Served in a leadership role requiring discipline, accountability, operational readiness, and team coordination.

AI, ENGINEERING INTELLIGENCE & INNOVATION

- Founder and architect of IntelligenceOS, a modular AI platform concept for reusable intelligence plugins across career, engineering, restaurant, and business operations.
- Built Career Intelligence as a system for resume management, job posting capture, application tracking, job analysis, ATS optimization, and interview preparation.
- Built Engineering Intelligence concepts and dashboards focused on production support visibility, feature intelligence, capacity forecasting, quality metrics, and executive reporting.
- Applying AI-assisted workflows to engineering leadership, operational triage, job search analytics, and software delivery improvement.

EDUCATION

Iowa State University - M.S. Computer Engineering | Ames, Iowa | GPA: 3.5

Campbell University - B.S. Computer Science | Buies Creek, North Carolina | GPA: 3.4

University of Texas at El Paso - Computer Science Coursework | El Paso, Texas

SELECTED TECHNICAL KEYWORDS

Java, Java/J2EE, Spring, REST APIs, SOAP, JSON, XML, JSP, Servlets, JavaBeans, Hibernate, SQL, Oracle, MySQL, AWS, Lambda, S3, DynamoDB, DocumentDB, Aurora, EC2, ECS, CloudWatch, Jenkins, Git, GitHub, GitLab, Docker, Tomcat, Linux, Unix, Python, Shell Scripting, C, C++, C#.NET, JavaScript, PHP, OAuth2, Swagger, JUnit, Maven, Angular, Redis, Jira, Confluence, DOORS, DO-178B, DO-254, Agile, SDLC, CI/CD, Production Support, Enterprise Architecture, Engineering Intelligence, AI-Assisted Development.